



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
PRE BOARD EXAMINATION

CLASS: X

SUBJECT: ROBOTICS & AI [SET A]

FULL MARKS: 100

TIME: 2 HOURS

Answers to this Paper must be written on the paper provided separately.

You will not be allowed to write during the first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

This paper consists of five printed pages. This Paper is divided into two Sections.

Attempt all questions from Section A and any four questions from Section B.

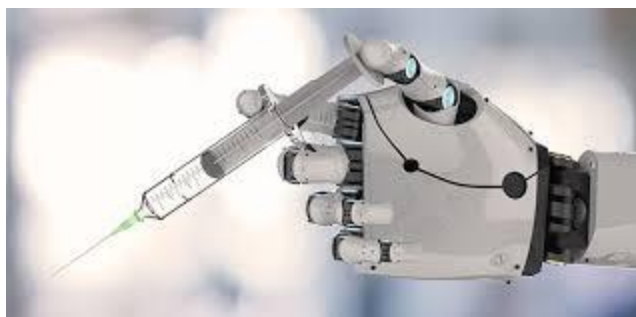
The intended marks for questions or parts of questions are given in brackets [].

Question 1

[20x1=20]

Choose the correct option. Do not copy the question; write only the correct answer.

(i) Identify the robot shown below :



(a) Agricultural robot (b) Medical robot (c) Entertainment robot (d) Warehouse robot

(ii) A **drone** delivering goods **autonomously** must use which combination of sensors for navigation?

- (a) Touch and sound (b) GPS and vision
(c) Temperature and force (d) Pressure and humidity

(iii) If the driver gear has 10 teeth and the driven gear has 40 teeth, the gear ratio is:

- (a) 1 : 4 (b) 4 : 1 (c) 2 : 1 (d) 1 : 2

(iv) The *stage* of the AI Project Framework that defines **4W's** of the problem is:

- (a) Problem Scoping (b) Data Acquisition (c) Modelling (d) Evaluation

- (v) Internal sensors in robots help to :
- (a) detect surrounding objects
 - (b) measure position, velocity & force
 - (c) process data
 - (d) control lights
- (vi) The component converting control signals into physical motion is called :
- (a) Controller
 - (b) Actuator
 - (c) Sensor
 - (d) Feedback
- (vii) Which statement about controllers is **most accurate**?
- (a) They sense environmental data.
 - (b) They supply power to the robot.
 - (c) They process inputs and issue commands to actuators.
 - (d) They replace sensors during testing.
- (viii) A **force sensor** is used when a robot must :
- (a) measure applied load or pressure
 - (b) detect color
 - (c) calculate distance
 - (d) track temperature
- (ix) Bevel gears are mainly used to :
- (a) change the direction of motion by 90°
 - (b) increase linear speed
 - (c) store torque energy
 - (d) cool motors
- (x) Arrange the correct control flow:
- (a) Sensor → Controller → Actuator → Feedback
 - (b) Actuator → Controller → Sensor → Feedback
 - (c) Controller → Actuator → Sensor → Feedback
 - (d) Sensor → Actuator → Controller → Feedback
- (xi) Cobots differ from industrial robots because they :
- (a) require isolation
 - (b) operate with human collaboration
 - (c) perform only welding
 - (d) lack safety sensors
- (xii) Which combination of actuators is used in a robot with both rotary & linear motion?
- (a) Servo and Pneumatic
 - (b) Stepper and Electric
 - (c) Linear and Rotary
 - (d) Rotary and Hydraulic
- (xiii) Increasing gear ratio results in :
- (a) higher torque and lower speed
 - (b) lower torque and higher speed
 - (c) constant torque
 - (d) equal torque and speed
- (xiv) Which of the following shows a **feedback-based** automatic control system?
- (a) Manual fan regulator
 - (b) Automatic room heater with thermostat
 - (c) Simple bulb switch
 - (d) Hand crank

(xv) **Assertion (A):** Sensors provide data to the controller.

Reason (R): Controllers are responsible for physical motion.

- (a) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (b) Both **A** and **R** true and **R** is not the correct explanation of **A**
- (c) **A** true, **R** false
- (d) **A** false, **R** true

(xvi) Which AI Project Framework stage involves checking how well the model performs?

- (a) Problem Scoping (b) Modelling (c) Evaluation (d) Deployment

(xvii) A **vision sensor** in a robot mainly converts :

- (a) sound → signal (b) image → digital data
- (c) force → voltage (d) light → temperature

(xviii) In **matplotlib**, which function displays a line chart?

- (a) plt.line() (b) plt.plot() (c) plt.graph() (d) plt.linechart()

(xix) To join elements of a list L into a single string, the function used is _____.

- (a) L.concat() (b) ' '.join(L) (c) L.combine() (d) join(L)

(xx) The most suitable **single-board computer** for controlling small robots in schools is :

- (a) Raspberry Pi (b) Hard Disk (c) SSD (d) Router

Question 2

[10x2=20]

(i) State *two* differences between **robots** and **machines**.

(ii) Write *any two* **functions of sensors** in a robot.

(iii) Define **gear ratio** and state how it affects torque and speed.

(iv) What is the purpose of **actuators**? Give one example.

(v) Mention *two* differences between **linear** and **rotary** actuators..

(vi) Mention *any two* components of the **AI Project Framework**.

(vii) List *any two* advantages of **Tinkercad** in visualizing robotic motion.

(viii) Find the **output** of the following code:

```
text = "Robotics"
print(text[2:7])
```

(ix) Find the **output** of the following code:

```
nums = [5, 10, 15, 20]
nums.append(25)
nums.insert(2, 12)
print(nums)
```

(x) Find the **output** of the following code:

```
tup = (2, 4, 6, 8)
mylist = list(tup)
mylist.remove(4)
tup = tuple(mylist)
print(len(tup))
```

SECTION B (60 Marks)

(Answer any four questions from this Section.)

The answers in this section should consist of the programs in either Python environment or any program environment with Python as the base. Each program should be written using variable description / mnemonic codes so that the logic of the program is clearly depicted. Flowcharts and algorithms are not required.

Question 3

- (i) Explain the **Problem Scoping** and **Data Acquisition** stages of the AI Project Framework with examples. [3]
- (ii) Describe any two **types of gears** and their specific applications in robots. [3]
- (iii) Using the given data, write a **Python program** to draw a **pie chart** of a robot's power consumption:

Movement – 40 %, Processing – 25 %, Sensors – 20 %, Communication – 15 %.

Label each section clearly and display the chart. [9]

Question 4

- (i) Differentiate between **internal** and **external** sensors and give one example of each. [3]
- (ii) Describe the **Modelling** and **Evaluation** stages in the AI Project Cycle. [3]
- (iii) The following data represents temperature readings at five time intervals:
Time=[1, 2, 3, 4, 5] , Temperature =[30, 32, 34, 33, 31]

Write a **Python program** using matplotlib to:

1. Create two lists for time and temperature,
2. Plot a **line graph** showing variation of temperature with time with *time* on x-axis and *temperature* on Y-axis,”
3. Add the appropriate title “temperature readings. [9]

Question 5

- (i) Define **actuators** and state two advantages of using rotary actuators. [3]
- (ii) Explain how **feedback** helps maintain accuracy in a robotic arm. [3]
- (iii) Write a **Python program** to perform the following tasks:
 1. Create a list of 5 integers. [1]
 2. Append a number to the list. [1]
 3. Sort the list in **descending order**. [1]
 4. Convert the list into a tuple. [1]

5. Access and display the 3rd element of the tuple. [1]
6. Convert the tuple back into a list and display it. [2]
7. Remove the last element of the list. [1]
8. Display the final list after conversion. [1]

Question 6

- (i) Explain how **gear ratio** influences the mechanical efficiency of a robot. [3]
- (ii) Discuss any two advantages of **Tinkercad** in designing robotic systems. [3]
- (iii) Write a **Python program** that:
 1. Takes a string input from the user. [1]
 2. Count the number of vowels in the string. [2]
 3. Check whether the string contains only alphabets. [1]
 4. Check if the string starts with a given character input by user. [1]
 5. Find the index of a character in the string input by user. [2]
 6. Replace all occurrences of a character with another both inputs by user and display the string. [2]

Question 7

- (i) Explain how **sensors, actuators, and controllers** coordinate to complete a robotic task. [3]
- (ii) Describe the role of **data evaluation** in improving AI model accuracy. [3]
- (iii) Write a **Python string program** using the given string:
text = "Robotics and Artificial Intelligence"
 Perform the following:
 1. Display characters from index 3 to 12. [1]
 2. Convert the entire string to uppercase. [1]
 3. Check if the string **ends with** "Intelligence". [1]
 4. Replace "Artificial" with "Smart". [2]
 5. Display the new string. [1]
 6. Find the length of the string and print it. [2]
 7. Check if 'r' exists in the string. [1]

Question 8

- (i) Describe the **steps involved in designing a robotic component** using Tinkercad. [3]
- (ii) Mention three common **data-visualization types** used in robotics and AI. [3]
- (iii) Using the given string:
line = "Python3 is fun"
 Write a **Python program** to:
 1. Print the length of the string. [1]
 2. Check if it is alphanumeric. [2]
 3. Convert it to lowercase and print it. [2]
 4. Count the occurrences of 'n' and print it. [2]
 5. Print substring from index 0 to 6 with a step of 2. [2]